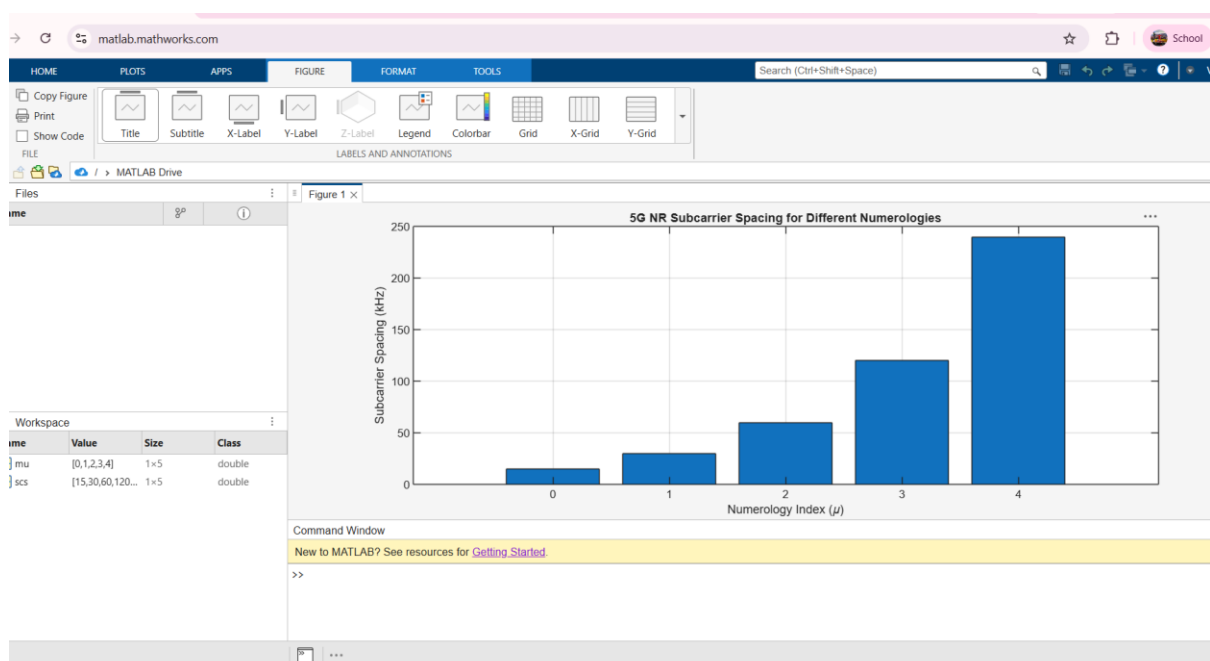


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EXP 6

```
Obj1 '  
clc;  
clear;  
close all;  
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)  
mu = 0:4; % Numerology Index  
bar(scs)  
grid on  
set(gca,'XTick',1:5)  
set(gca,'XTickLabel',mu)  
xlabel('Numerology Index (\mu)')  
ylabel('Subcarrier Spacing (kHz)')  
title('5G NR Subcarrier Spacing for Different Numerologies')
```



Obj 2

```
clear;
```

```
close all;
```

```
clc;
```

```
% Subcarrier Spacing (kHz)
```

```
scs = [15 30 60 120 240];
```

```
% Slot Duration (ms)
```

```
slot = [1 0.5 0.25 0.125 0.0625];
```

```
% Plot
```

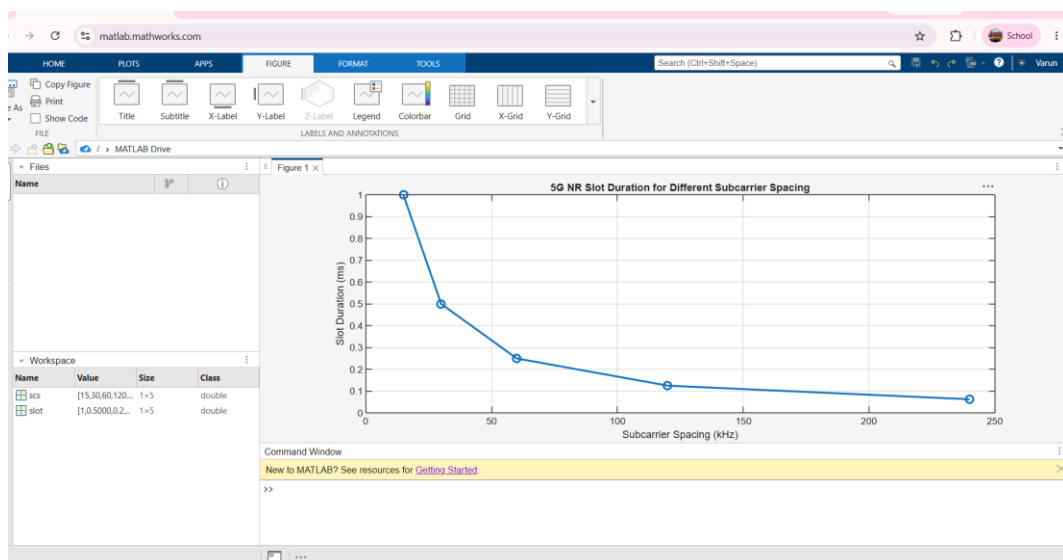
```
plot(scs, slot, 'o-', 'LineWidth', 2, 'MarkerSize', 8);
```

```
grid on;
```

```
xlabel('Subcarrier Spacing (kHz)');
```

```
ylabel('Slot Duration (ms)');
```

```
title('5G NR Slot Duration for Different Subcarrier Spacing');
```



Obj3

close all;

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
```

figure

```
bar(scs, symbols)
```

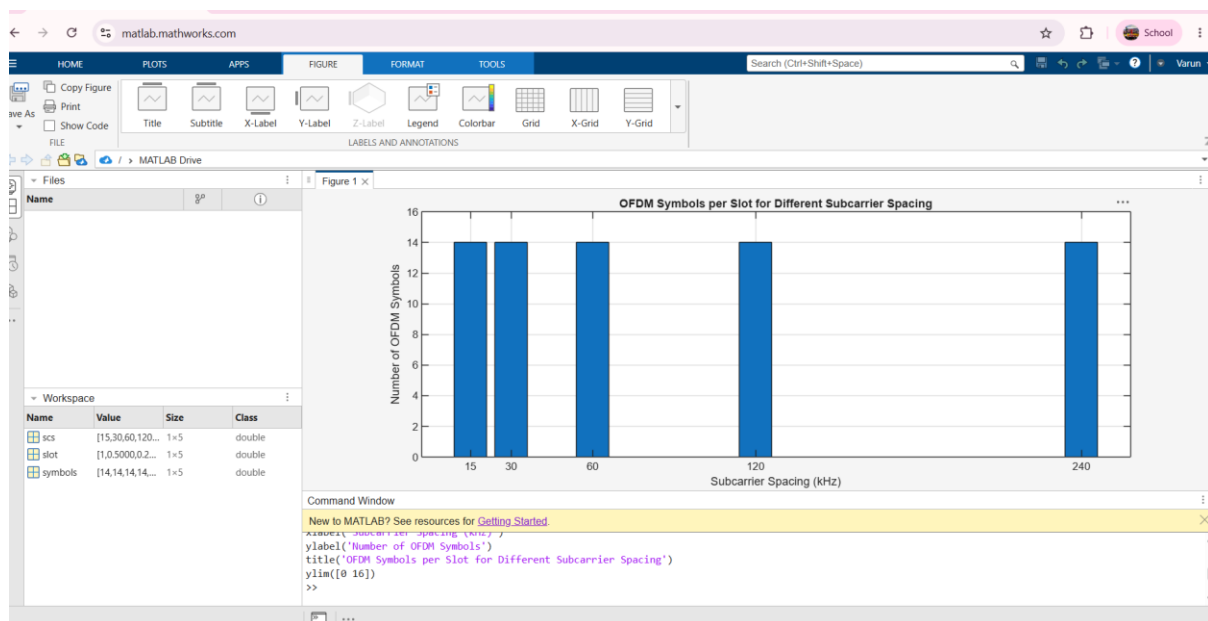
grid on

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Number of OFDM Symbols')
```

```
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
```

```
ylim([0 16])
```



Experiment 7

Obj 1

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Beamforming Gain (dB)
```

```
figure
```

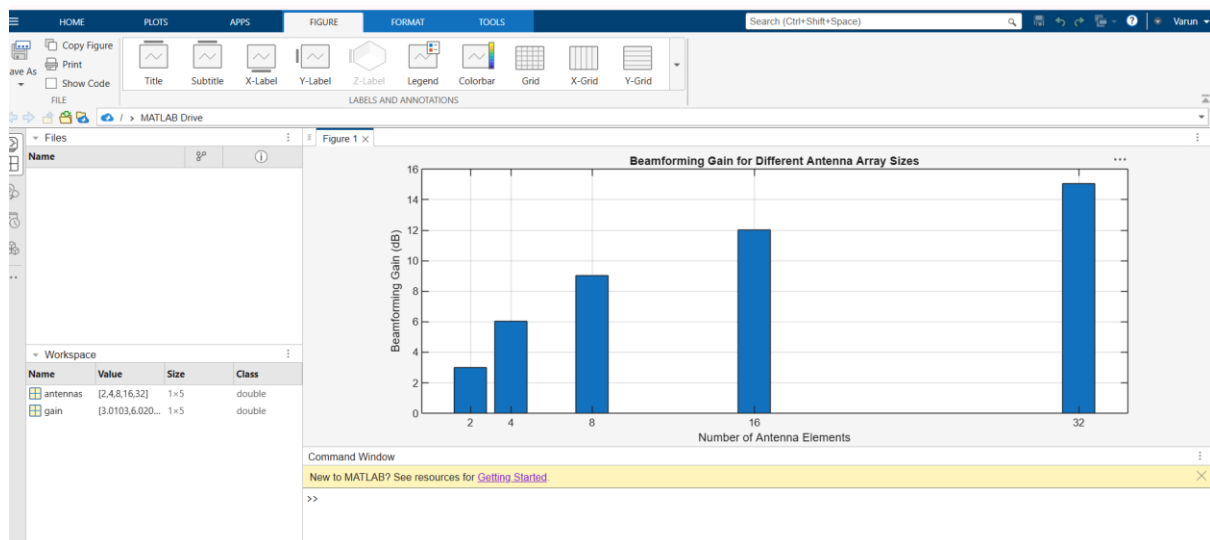
```
bar(antennas, gain)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beamforming Gain (dB)')
```

```
title('Beamforming Gain for Different Antenna Array Sizes')
```



Obj 2

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
```

```
figure
```

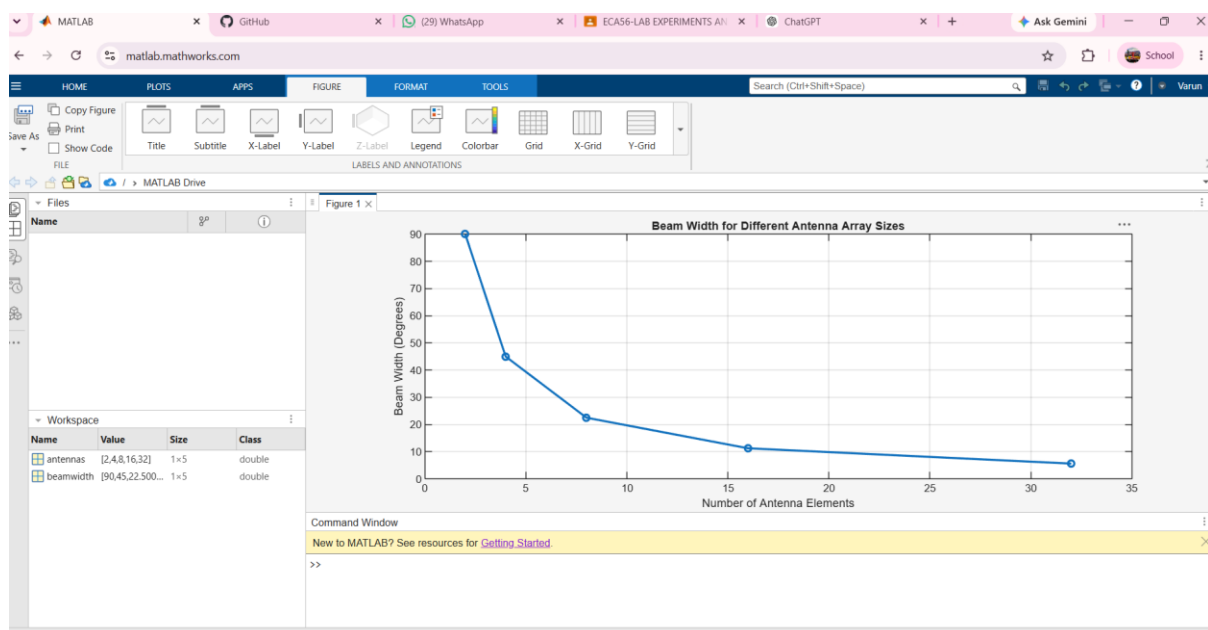
```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('Beam Width for Different Antenna Array Sizes')
```



Obj 3

```
clc;
```

```
clear;
```

```
close all;
```

```
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
```

```
gain = [1 1 1 1 1]; % Constant normalized gain
```

```
figure
```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
```

```
title('Main Lobe Direction in Analog Beamforming')
```

